Dynamic Optimal Pricing with Love of Variety

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Introduction

Research question

How do firms augment their optimal pricing strategy when love of variety is present?

- Repeated consumption may reduce the appeal of a product.
- Purchase histories can reveal when a consumer is ready to switch.
- Retailers can leverage that information to optimally price according to the distribution of consumers.
  - Dynamic price setting
  - Targeting

Motivation

Motivation

- **Pricing:** knowing how much consumers value variety helps firms tailor prices. In the model, this raises profit by about \$0.01 per consumer over two periods.
- **Demand:** flavor preferences change little when fatigue is included; some fatigue parameters are weakly identified.

Contribution

- Estimate distribution of LOV types
  - Lattin (1987), Chintagunta (1996), Dubé (2003)
- Connect switching preferences and state dependence to the information a retailer uses for pricing
  - Dubé and Misra (2023), Moorthy and Tehrani (2023), Karle and Reisinger (2024)

Consumer problem: switching in both periods

$r \in \{H, L\}$ is type; $t \in \{1, 2\}$ is period; $\ell \in \{j, k\}$ indexes products.
$h_0 = a \in \{j, k, 0\}$ is the observed initial state; h_1 is the first choice. State 0 means no prior product.

$$U_{r\ell t} = \underbrace{\beta_\ell}_{\text{baseline utility}} + \underbrace{\gamma_r \mathbf{1}\{h_{t-1} \neq 0, \ell \neq h_{t-1}\}}_{\text{switching reward}} + \underbrace{\alpha p_{\ell t}}_{\text{price effect}} + \varepsilon_{\ell t}, \qquad U_{r0t} = \varepsilon_{0t}.$$

- $\gamma_H > \gamma_L$: H values switching more; $\alpha < 0$ is the common price coefficient.
- $p_{\ell 1}(a)$ and $p_{\ell 2}(a, h_1)$ are freely chosen transaction prices.
- ε are independent type-I extreme-value shocks; $V_{rlt} = U_{rlt} - \varepsilon_{lt}$ is deterministic utility.

$$s_{r\ell t} = \Pr(h_t = \ell \mid r, \text{history}) = \frac{e^{V_{r\ell t}}}{1 + \sum_{m \in \{j,k\}} e^{V_{rmt}}}; \quad s_{r0t} = 1 - s_{rjt} - s_{rkt}.$$

Consumers maximize current utility. Initial states are equally likely and independent of type; a period-1 outside choice exits permanently.

Producer problem: induce, then harvest

$P = \{p_{\ell 1}(a), p_{\ell 2}(a,h)\}$ is the full price menu; c_ℓ is unit cost.
$\mu_a = \Pr(h_0 = a) = 1/3$, $q = \Pr(H)$, and $\delta \in (0,1]$ discounts firm profit.

$$\underbrace{w_r(a,h)}_{\text{terminal profit after first purchase } h} = \sum_{\ell \in \{j,k\}} s_{r\ell 2}(a,h) \underbrace{[p_{\ell 2}(a,h) - c_\ell]}_{\text{unit margin}},$$

$$\underbrace{\pi_r(P)}_{\text{value per arriving type-}r\text{ consumer}} = \sum_{a \in \{j,k,0\}} \mu_a \sum_{h \in \{j,k\}} s_{rh1}(a) \left[p_{h1}(a) - c_h + \delta w_r(a,h)\right].$$

$$P^*(q) \in \arg\max_P \{q\pi_H(P) + (1-q)\pi_L(P)\}.$$

- Common menu: type is unobserved; prices depend on observed history. Observed type: optimize a separate menu for each r.
- Commit before choices; account for type sorting into histories. Period-1 prices affect current earnings and future profit.

Optimal prices when type is known

For type r, c_ℓ is unit cost, $\alpha < 0$ is the price coefficient, and s_{r0t} is the outside share. The initial state is a; the first purchase is h.

Period 2: harvest

$$p^*_{r\ell 2}(a,h) - c_\ell = -\frac{1}{\alpha s^*_{r02}(a,h)}, \qquad \ell \in \{j,k\}.$$

Terminal unit margins are equal across products at each history.

Period 1: induce with continuation value

$$p^*_{r\ell 1}(a) - c_\ell + \delta w^*_r(a,\ell) = -\frac{1}{\alpha s^*_{r01}(a)}, \qquad \ell \in \{j,k\}.$$

$w^*_r(a,\ell)$ is optimal period-2 profit after buying ℓ; δ is the firm discount factor.
Larger continuation value supports a lower current margin, holding the right-hand side fixed.

Interior conditions; shares and continuation values are evaluated at optimal prices.

Unobserved type

Free pricing: what changes?

Illustrative calibration

$\beta = (2.2, 2.0)$, $c = (1.2, 0.8)$, $\alpha = -1.5$, $\delta = 0.95$.
Baseline $(\gamma_H, \gamma_L) = (2, 1)$; initial states $\mu_j = \mu_k = \mu_0 = 1/3$.

- Optimize all product prices separately by period and observed history.
- For known type, interior terminal prices imply equal product margins at each history.
- Measure induction by current profit sacrificed relative to the one-period optimum; measure harvesting by period-2 profit.

These are model illustrations, not estimated customer values. Pricing conditions

Optimal prices depend on the initial state

History	Observed H		Observed L		Common 50/50	
	p_j	p_k	p_j	p_k	p_j	p_k
Period 1: initial state						
j	2.464	2.176	2.230	1.886	2.311	2.057
k	2.383	2.095	2.172	1.829	2.303	1.926
0	1.816	1.528	1.970	1.627	1.897	1.578
Period 2: state transition						
$j \to j$	2.859	2.459	2.540	2.140	2.616	2.288
$j \to k$	2.741	2.341	2.480	2.080	2.671	2.210
$k \to j$	2.859	2.459	2.540	2.140	2.695	2.371
$k \to k$	2.741	2.341	2.480	2.080	2.613	2.152
$0 \to j$	2.859	2.459	2.540	2.140	2.653	2.332
$0 \to k$	2.741	2.341	2.480	2.080	2.644	2.180

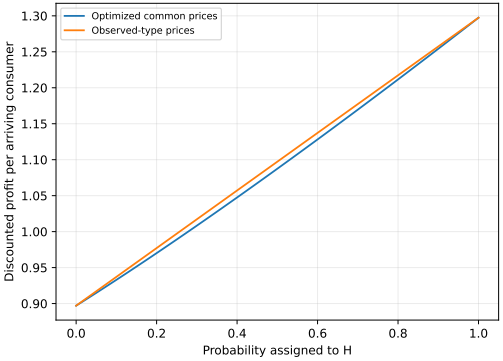
All prices shown. j, k: previous product; 0: no prior product. Common menu uses $q = 0.5$.

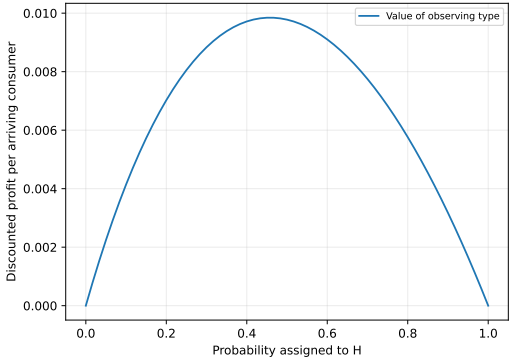
Period-1 induction and period-2 harvesting: results

Prices	True type	Period 1	$\delta\times$ Period 2	Total
Common	H	0.7045	0.5848	1.2892
Common	L	0.5592	0.3265	0.8857
Observed type	H	0.7215	0.5759	1.2974
Observed type	L	0.5499	0.3471	0.8970

- Profit is per arriving consumer, averaged over all initial states and choice shocks.
- Total equals period-1 profit plus discounted period-2 profit.

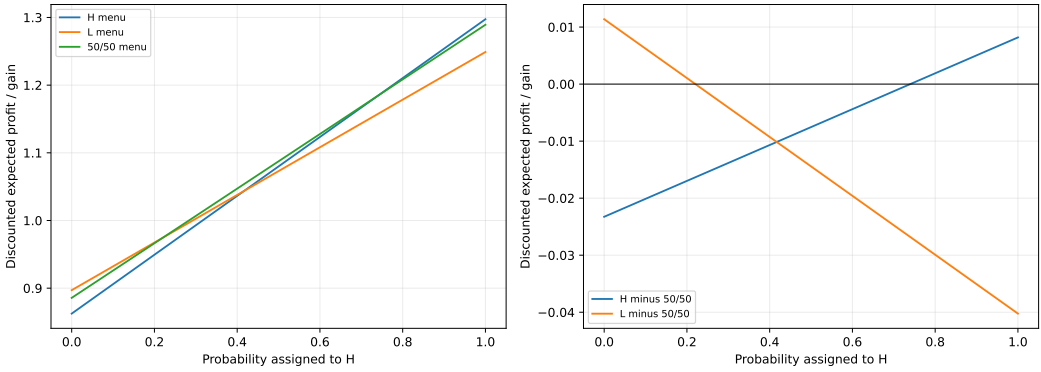
Profits and the value of type information





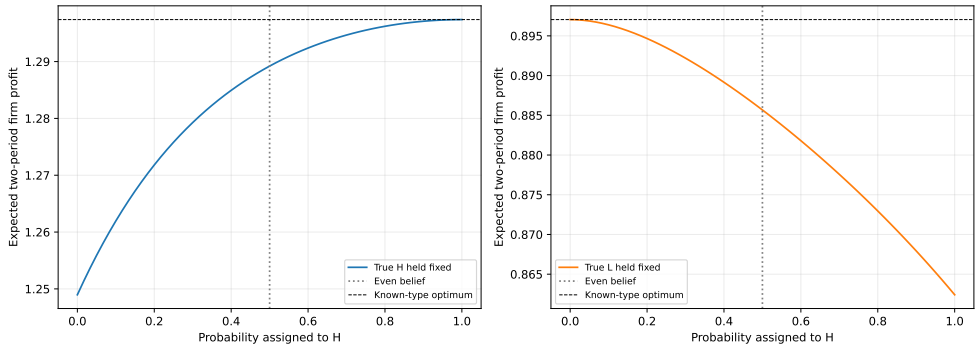
At $q = 0.5$: common profit 1.08744, observed-type profit 1.09722, gain 0.00978.

Choosing among three fixed price menus



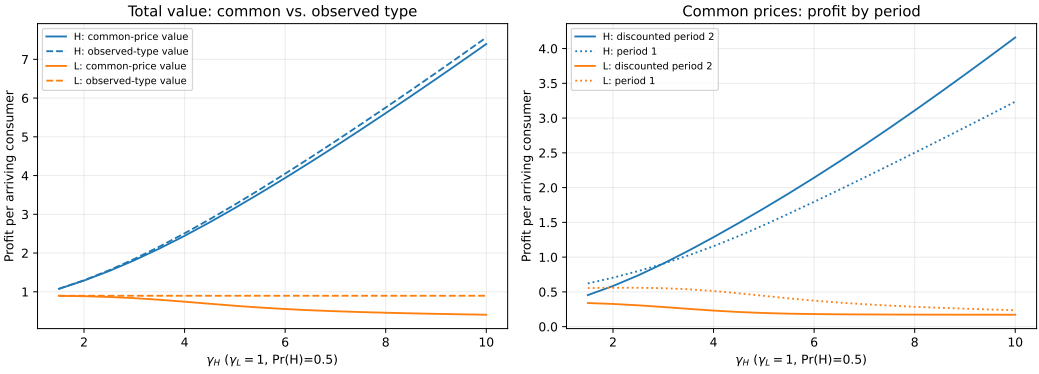
H, L, and 50/50 benchmark menus remain fixed as q varies. Freely reoptimizing prices weakly dominates these three choices.

Mispricing: the true type stays fixed



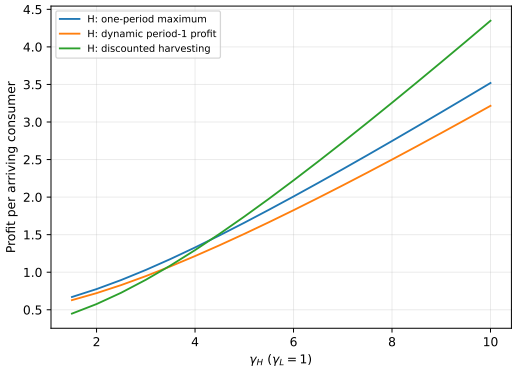
Customer value $\pi_r(P) = \mathbb{E}[\text{period-1 profit} \mid r] + \delta\mathbb{E}[\text{period-2 profit} \mid r]$, $\delta = 0.95$.
Firm profit per arriving consumer; averages over initial states and choices. Period-1 exit earns zero thereafter.
Both panels use $q = \Pr(H)$: correct certainty is $q = 1$ for H and $q = 0$ for L.

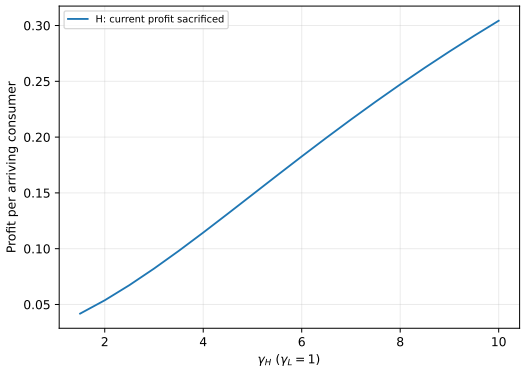
Who is most valuable as γ_H changes?



Hold $\gamma_L = 1$, $q = 0.5$; reoptimize at $\gamma_H = 1.5, 2, \ldots, 10$ (steps of 0.5).
Left: total value by information regime. Right: period contributions under common prices.

Induction in period 1 and harvesting in period 2





Known-H benchmark: induction sacrifice is one-period optimal profit minus dynamic period-1 profit. Harvesting includes survival and is discounted by δ.

Data

- Nielsen HomeScan and MarketScan, 2022–2024.
- Eight markets: Atlanta, Chicago, Denver, Houston, Des Moines, Phoenix, Philadelphia, San Diego.
- Single-person households reduce ambiguity about whose tastes drive purchases.
- Single-serving yogurt offers flavor variation and limited storability.

Sample

	Full sample	Estimation sample
Households	5,408	998
Mean store trips / household	308.15	400.05
Mean observed yogurt choice weeks	—	3.91
Coupon / deal purchase rate	26.52%	23.10%
Mean weekly yogurt units (purchasers)	—	8.44

Implication

The panel contains limited yogurt history for many households.

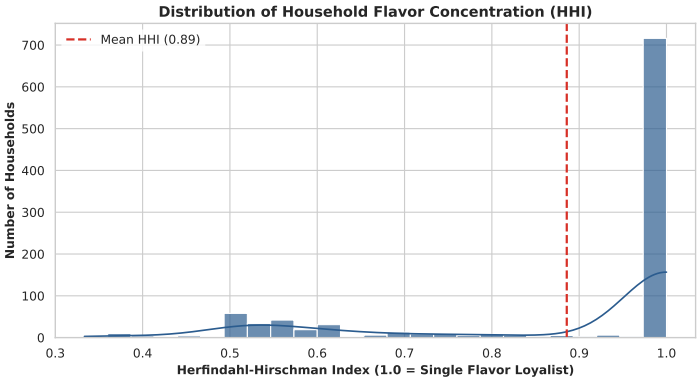
Flavor groups

- **Berry:** berry-labeled flavors plus cherry.
- **Plain:** a separate category.
- **Other:** all remaining flavors.
- **Outside option:** no yogurt purchase.

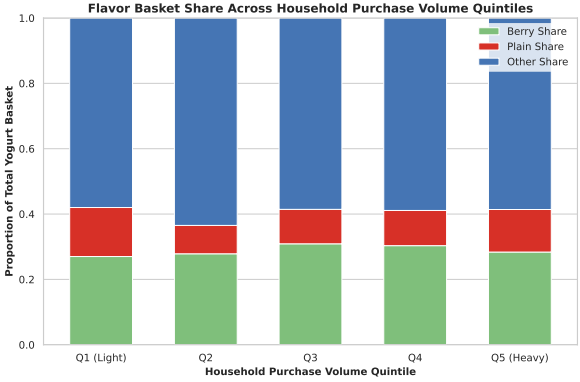
Tradeoff

Aggregation makes estimation tractable but hides switching within a flavor group.

Flavor concentration



Flavor shares



Demand model

$$U_{ijt} = \underbrace{\beta_0 + \beta_i' X_{jt}}_{\text{baseline tastes}} + \underbrace{\gamma_i \Xi_{ijt}}_{\text{history state}} + \underbrace{\alpha p_{jt}}_{\text{price}} + \varepsilon_{ijt}.$$

$$\text{s.t. } \Xi_{ijt} = \lambda \sum_{s=1}^{8} \delta^{t-s} \Xi_{ij,t-s} + (1-\lambda) \sum_{k \neq j} \rho_k X_{k,t-1}$$

- Inside choices: berry, plain, other; outside utility normalized to zero.
- Ξ_{ijt} summarizes prior consumption and switching.
- Negative fatigue coefficients reduce utility as the history index rises.

Simulation procedure Simulation results Utility specification

Identification

- **Stable tastes:** persistent differences in flavor choices.
- **Fatigue:** changes in choices associated with a household's own history.
- **Challenge:** heterogeneity, prices, and unobserved shocks can also generate switching.
- **Limited history and coarse groups** make the distinction difficult.

Price control function

$$p_{jmt} = a + b \underbrace{\bar{p}_{j,-m,t}}_{\text{other-market price}} + \tau_t + \eta_{jmt}, \qquad \hat{\eta}_{jmt} = p_{jmt} - \hat{p}_{jmt}.$$

1. Match household choices to flavor-specific prices and purchase histories.
2. Regress local prices on other-market prices and time controls.
3. Include the fitted residual in utility to account for endogenous prices.

Required exclusion

Other-market prices must shift local prices without sharing unobserved local demand shocks, conditional on the controls.

Specifications

	No fatigue	With fatigue
Standard logit	Fixed β; $\gamma = 0$	Fixed β, γ
Random coefficient logit	Heterogeneous β; $\gamma = 0$	Heterogeneous β, γ

- Price and the control-function residual enter all four specifications.
- Compare baseline flavor tastes and price effects across columns.
- Assess fatigue means and dispersion separately from optimizer convergence.
- Identification depends on within-household history variation, valid price instruments, and the treatment of initial histories.

Panel likelihood

Demand estimates

	Standard logit		Random coefficients	
	No Variety	Variety	No Variety	Variety
Berry preference	−.7635	−.7576	−.7632	−.7489
Plain preference	−1.3899	−1.3897	−1.3930	−1.3956
Price	−.9836	−.9821	−.9836	−.9804
Control function	.1670	.1667	.1654	.1662

- All estimates relative to other flavors and the outside option

Main empirical result

The largest baseline flavor change is about 1.9% for berry in the random coefficient model.

LOV coefficients

	Standard logit	Random coefficients
Berry LOV	−.7262	−7.2030
Standard error	(.1485)	(1.5424)
Plain LOV	−.0351	−.0555
Standard error	(.1009)	(.1032)

- Berry has a negative estimated history coefficient.
- Plain is not statistically distinguishable from zero.
- Large changes across specifications warrant caution.

Heterogeneity

Random coefficient SD	Estimate	SE
β_{berry}	1.0513	.0916
β_{plain}	1.0513	.1492
γ_{berry}	148.4132	155.3868
γ_{plain}	1.0513	.1641

Diagnostic

The berry LOV dispersion is extremely large and imprecisely estimated.

Optimization success alone does not establish reliable identification.

Implied WTP

Magnitude in dollars	Standard	Random coefficients
Berry preference	.77	.76
Plain preference	1.42	1.42
Berry fatigue	.74	7.35
Plain fatigue	.04	.06

$$\text{Implied WTP} = \frac{|\text{utility coefficient}|}{|\alpha|}.$$

Interpretation

The large \$7.35 number belongs to berry LOV and is sensitive to the specification.

Conclusion

- Restricting type information can reduce retailer profits in this model.
- Type information changes optimal induction and harvesting prices.
- Attribute fatigue lowers consumer utility
- Difficult to disentangle preferences from love of variety

Thank You!

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Optimal prices when type is unobserved

Let $\omega_H = q$, $\omega_L = 1 - q$, $m_{\ell t} = p_{\ell t} - c_\ell$, and

$$M_{r\ell}(a) = m_{\ell 1}(a) + \delta w_r(a, \ell), \qquad v_r(a) = \sum_{h \in \{j,k\}} s_{rh1}(a) M_{rh}(a).$$

Period 1: current margin plus continuation value

$$\sum_{r \in \{H,L\}} \omega_r s_{r\ell 1}(a) \left[1 + \alpha\{M_{r\ell}(a) - v_r(a)\}\right] = 0.$$

Period 2: harvest from the consumers reaching history (a, h)

$$\sum_{r \in \{H,L\}} \omega_r s_{rh1}(a) s_{r\ell 2}(a, h) \left[1 + \alpha\{m_{\ell 2}(a, h) - w_r(a, h)\}\right] = 0.$$

Conditions hold for each product and observed history. In period 2, $\omega_r s_{rh1}(a)$ accounts for type selection into the first purchase.

◂ Known-type conditions

Pricing exercise: parameters and timing

- $\beta = (2.2, 2.0)$, $c = (1.2, 0.8)$, $\alpha = -1.5$, $\delta = 0.95$.
- Initial states $j, k, 0$ have equal probabilities, independent of type; the firm observes the state.
- Baseline $\gamma_H = 2$, $\gamma_L = 1$; sensitivity varies γ_H from 1.5 to 10 in steps of 0.5 with $q = 0.5$.
- Commit to free prices by observed history. Consumers choose myopically; first-period outside choices exit.
- Three unconstrained optimizer starts per initial state, analytic gradients, and numerical stationarity checks.

◂ Profits

Free-price conditions: dynamic induction

For known type, define $w_r(h)$ as terminal profit and s_{r0t} as the outside share. At an interior optimum:

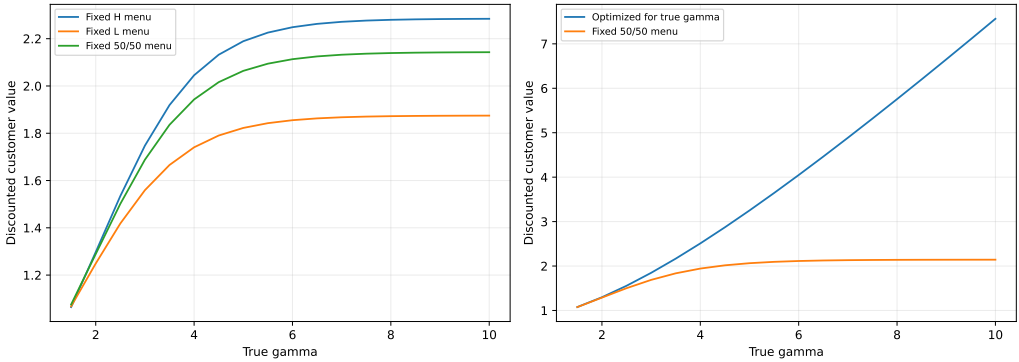
$$p_{r\ell 2}(h) - c_\ell = -\frac{1}{\alpha s_{r02}(h)},$$

$$p_{rh1}(a) - c_h + \delta w_r(a, h) = -\frac{1}{\alpha s_{r01}(a)}.$$

- Terminal margins are equal across products for each known type and history.
- Larger continuation value lowers the optimal first-period margin, all else equal.
- With unobserved heterogeneous types, optimize the weighted full objective; known-type formulas cannot be applied to pooled shares.

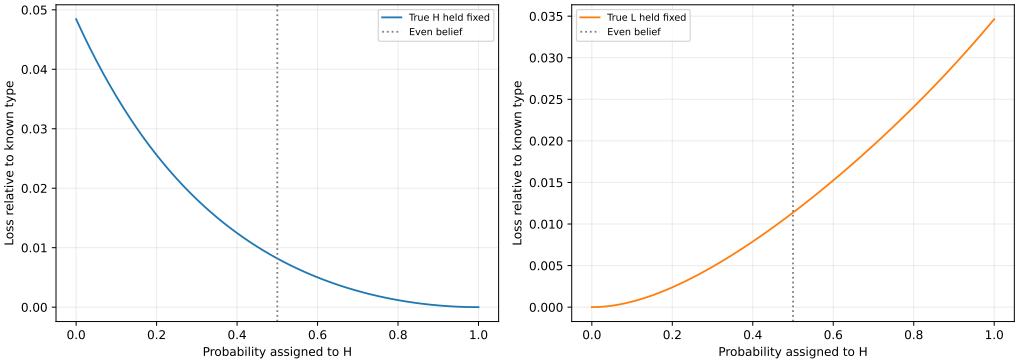
◂ Producer problem

Profit conditional on the true switching preference



This varies the actual preference under fixed benchmark menus or known-type optimized prices. It differs from varying γ_H while reoptimizing a two-type common menu.

The cost of choosing the wrong prices



Even beliefs: H loss 0.00818, L loss 0.01137.
Wrong certainty: H loss 0.04842, L loss 0.03464. Both axes show Pr(H).

Simulation procedure

Simulation design	Assigned value
Replications / evaluation periods / burn-in	1,000 / 100 / 5
Product attributes / baseline taste	$X_j \in \{1, 2, 3, 4, 5\}$; $\beta = 2$
Variety grid / choice shocks	$\gamma \in \{0, 3, 6, 9\}$; $\varepsilon_{jt} \sim \mathrm{T1EV}$

1. For each γ, initialize consumption histories during burn-in.
2. At each evaluation date, combine baseline attribute utility with the history-dependent distance reward and a choice shock.
3. Record the selected product, update its consumption history, and repeat.
4. Summarize choice probabilities across the simulated runs.

Purpose: show how variety changes choices in a controlled setting.

Simulation results ◂ Demand model

Simulation results

	$\gamma = 0$	$\gamma = 3$	$\gamma = 6$	$\gamma = 9$
Product 1 ($X = 1$)	.0016	.1352	.3321	.3973
Product 2 ($X = 2$)	.0084	.1325	.0541	.0084
Product 3 ($X = 3$)	.0437	.0844	.0026	.0000
Product 4 ($X = 4$)	.1934	.0553	.0023	.0002
Product 5 ($X = 5$)	.7529	.5926	.6089	.5941

Interpretation

Without variety, the highest-attribute product dominates. With stronger variety, choices concentrate at the endpoints.

This motivates modeling history; it does not establish identification in the yogurt panel.

Simulation procedure ◂ Demand model

Utility specification

For $j \in \{B, P, O\}$ (berry, plain, other), let $D_j^f = \mathbf{1}\{j = f\}$:

$$
\begin{aligned}
U_{ijt} &= V_{ijt} + \varepsilon_{ijt}, \\
V_{ijt} &= \beta_0 + \beta_i^B D_j^B + \beta_i^P D_j^P \\
&\quad + \gamma_i^B D_j^B \Xi_{ijt} + \gamma_i^P D_j^P \Xi_{ijt} + \alpha p_{j,m(i),t} + \sigma \hat{\eta}_{j,m(i),t}.
\end{aligned}
$$

- Other flavor is the reference; outside deterministic utility is zero.
- Ξ_{ijt} summarizes own past consumption and alternative-flavor history.
- History weight $\lambda = 0.7$ and decay $\delta = 0.9$ are fixed.
- Random coefficients allow persistent household differences in tastes and fatigue.

◂ Demand model

Panel likelihood

Conditional on household coefficients $b_i = (\beta_i^B, \beta_i^P, \gamma_i^B, \gamma_i^P)$:

$$P_{ijt}(b_i) = \frac{e^{V_{ijt}(b_i)}}{1 + \sum_{k \in \{B,P,O\}} e^{V_{ikt}(b_i)}}, \qquad P_{i0t}(b_i) = \frac{1}{1 + \sum_k e^{V_{ikt}(b_i)}}.$$

$$L_i(b_i) = \prod_{t \in \mathcal{T}_i} \prod_{j \in \{0,B,P,O\}} P_{ijt}(b_i)^{y_{ijt}}, \qquad y_{ijt} = \mathbf{1}\{\text{observed choice is } j\}.$$

$$\boxed{\hat{\theta} = \arg\max_\theta \sum_i \log\left[\int L_i(b)\, dF(b;\theta)\right]}$$

Simulation approximation: $\int L_i(b) dF(b;\theta) \approx R^{-1} \sum_{r=1}^R L_i(b_i^{(r)})$.

Maximize log-likelihood: $\ell(\theta) = \sum_{i,t,j} y_{ijt} \log P_{ijt}(\theta)$.

◂ Specifications

Discussion

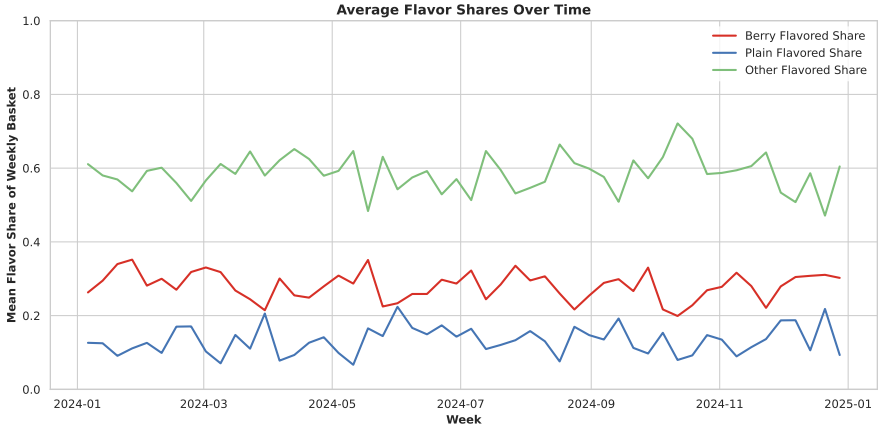
Questions for discussion

How can fatigue be distinguished from persistent unobserved tastes?

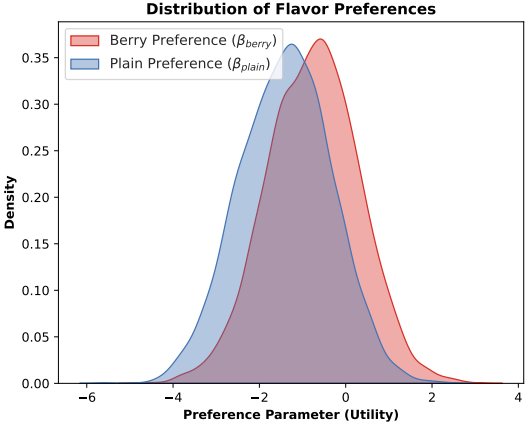
Which purchase histories are most informative about switching types?

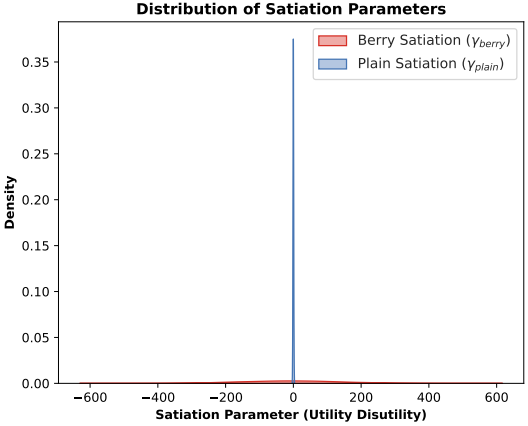
What consumer-welfare evidence is needed to evaluate privacy restrictions?

Flavor shares over time

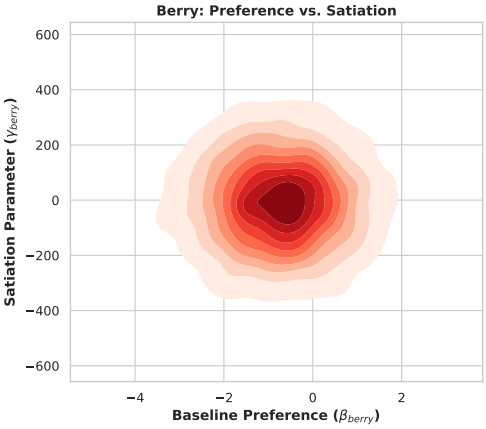


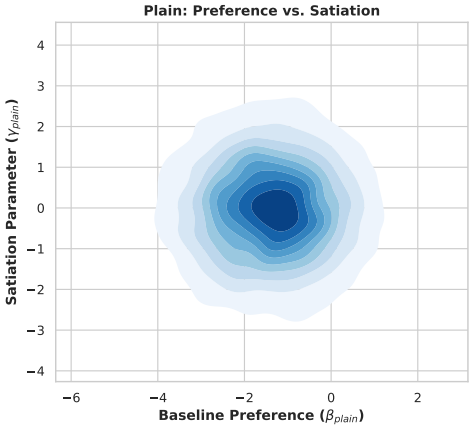
Marginal distributions





Joint distributions





References

- Attribute satiation: McAlister; Lattin (1987).
- Variety and brand choice: Givon (1984); Chintagunta (1999).
- Bundles and variety: Dubé (2004).
- Price control functions: Petrin and Train (2010).
- Personalized pricing: Dubé and Misra (2023).

Full bibliographic entries appear in the accompanying paper.